

Primary 5 Mathematics Practice Sheets

Area of Triangles, Composite Figures and Remainder Concept

5 sets x 10 questions | A4 printer-friendly | Maximum 2 questions per page

Instructions:

1. Show your working clearly for every question.
2. Use the formula: Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$.
3. For remainder-concept questions, draw a model or use branching to track the remainder.
4. Check units carefully: cm^2 , m^2 , litres, dollars, or number of items.

Note: These are original practice questions based on the question types provided, aligned to Primary 5 practice needs.

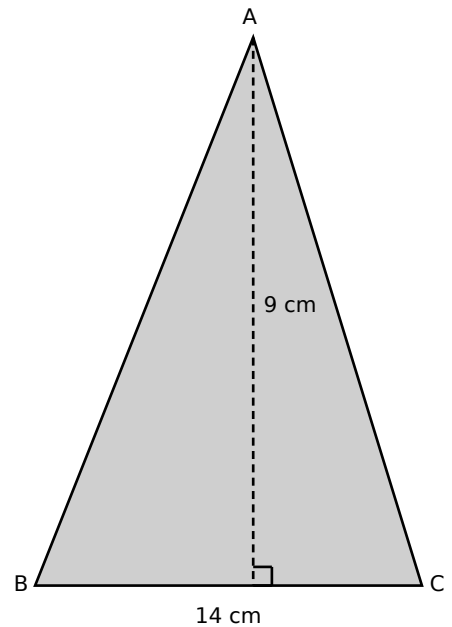
Set 1 - Area and Remainder

Level: Easy to Medium

Name: _____ Class: _____ Date: _____

Page 1

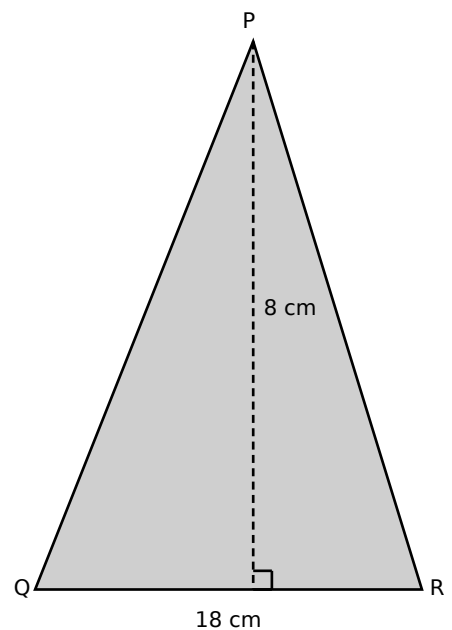
- 1.** In Triangle ABC, base BC is 14 cm and the perpendicular height AD is 9 cm. Find the area of Triangle ABC.



Working:

Answer:

- 2.** In Triangle PQR, QR is the base. QR = 18 cm and the perpendicular height from P to QR is 8 cm. Find the area of Triangle PQR.



Working:

Answer:

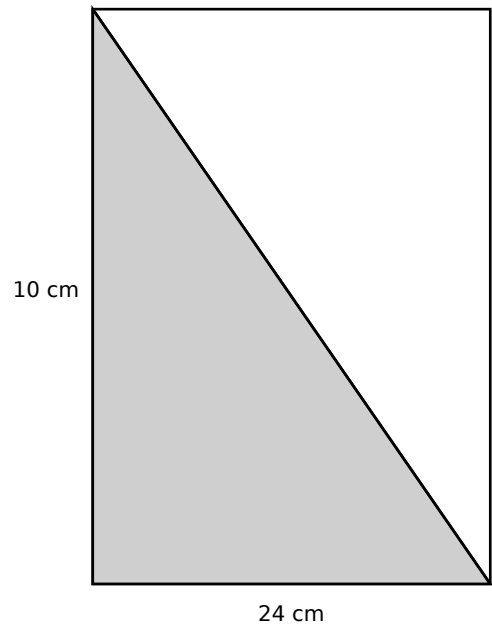
Set 1 - Area and Remainder

Level: Easy to Medium

Name: _____ Class: _____ Date: _____

Page 2

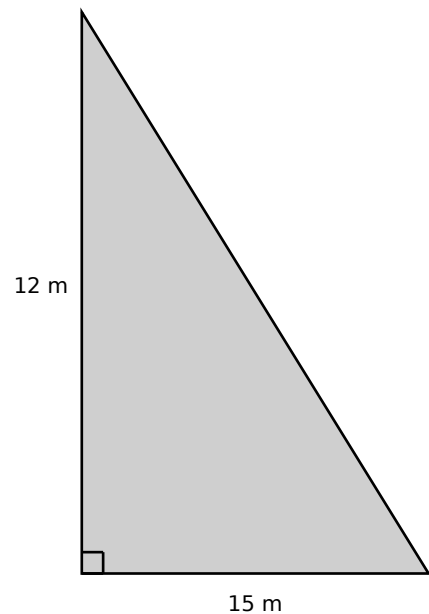
- 3.** A rectangle measures 24 cm by 10 cm. A diagonal divides it into two equal triangles. Find the area of one shaded triangle.



Working:

Answer:

- 4.** A right-angled triangle has perpendicular sides of 15 m and 12 m. Find its area.



Working:

Answer:

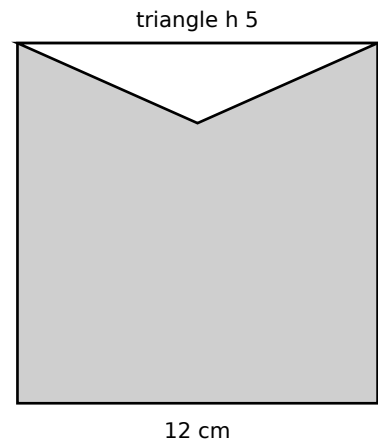
Set 1 - Area and Remainder

Level: Easy to Medium

Name: _____ Class: _____ Date: _____

Page 3

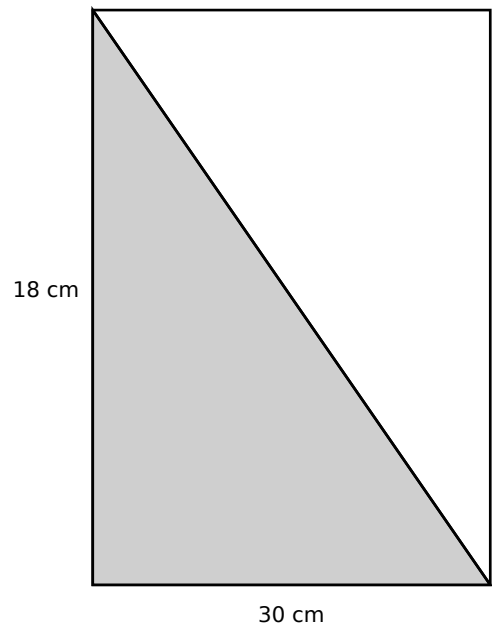
5. A square has side 12 cm. An unshaded triangle at the top has base 12 cm and height 5 cm. Find the shaded area.



Working:

Answer:

6. A rectangle measures 30 cm by 18 cm. A diagonal divides it into two equal triangles. Find the area of the shaded triangle.



Working:

Answer:

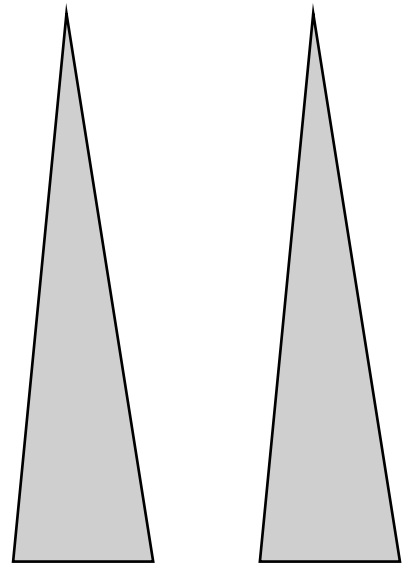
Set 1 - Area and Remainder

Level: Easy to Medium

Name: _____ Class: _____ Date: _____

Page 4

7. Two identical shaded triangles each have base 8 cm and height 6 cm. Find the total shaded area.

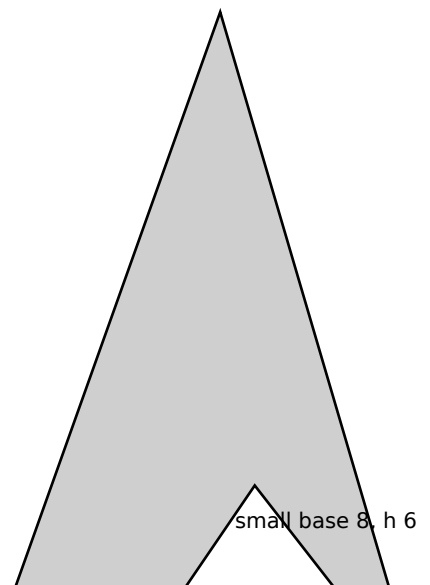


each: base 8 cm, height 6 cm

Working:

Answer:

8. A large triangle has base 20 cm and height 12 cm. A small unshaded triangle inside it has base 8 cm and height 6 cm. Find the shaded area.



large triangle: base 20, h 12

small base 8, h 6

Working:

Answer:

Set 1 - Area and Remainder

Level: Easy to Medium

Name: _____ Class: _____ Date: _____

Page 5

- 9.** Sara had some money. She spent $\frac{1}{4}$ of it on a book and then $\frac{1}{3}$ of the remainder on a pen. She had \$60 left. How much money did she have at first?

Working:

Answer:

- 10.** A tank had 80 litres of water. $\frac{1}{5}$ of the water was used in the morning and $\frac{1}{4}$ of the remaining water was used in the afternoon. How much water was left?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	63 cm ²	$\frac{1}{2} \times 14 \times 9 = 63 \text{ cm}^2$
2	72 cm ²	$\frac{1}{2} \times 18 \times 8 = 72 \text{ cm}^2$
3	120 cm ²	$\frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$
4	90 m ²	$\frac{1}{2} \times 15 \times 12 = 90 \text{ m}^2$
5	114 cm ²	Square $12 \times 12 = 144$; triangle $= \frac{1}{2} \times 12 \times 5 = 30$; shaded $= 114 \text{ cm}^2$
6	270 cm ²	$\frac{1}{2} \times 30 \times 18 = 270 \text{ cm}^2$
7	48 cm ²	Each triangle $= \frac{1}{2} \times 8 \times 6 = 24$; total $= 48 \text{ cm}^2$
8	96 cm ²	Large $= 120$; small $= 24$; shaded $= 96 \text{ cm}^2$
9	\$120	Left $= \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ of total. $\frac{1}{2} = \$60$, so total $= \$120$
10	48 litres	After morning: $80 \times \frac{4}{5} = 64$. Afternoon used $\frac{1}{4} \times 64 = 16$. Left $= 48$ litres

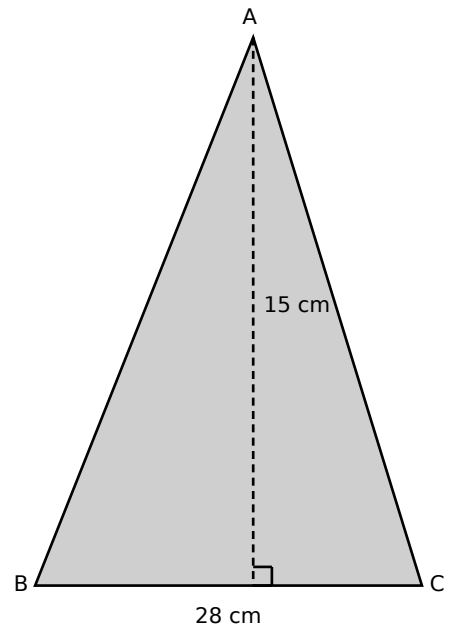
Set 2 - Shaded Areas

Level: Medium

Name: _____ Class: _____ Date: _____

Page 1

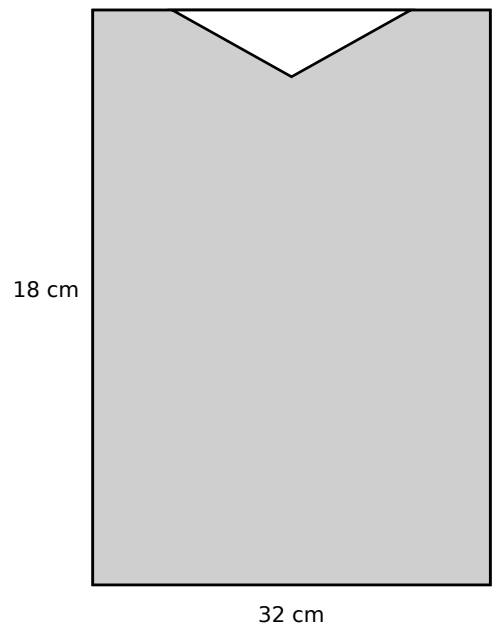
1. A shaded triangle has base 28 cm and perpendicular height 15 cm. Find its area.



Working:

Answer:

2. A rectangle measures 32 cm by 18 cm. An unshaded triangle at the top has base 20 cm and height 7 cm. Find the shaded area of the rectangle.



Working:

Answer:

Set 2 - Shaded Areas

Level: Medium

Name: _____ Class: _____ Date: _____

Page 2

3. A rectangular paper is folded at one corner. The triangular folded part has base 6 cm and height 8 cm. Find the area of the folded part.

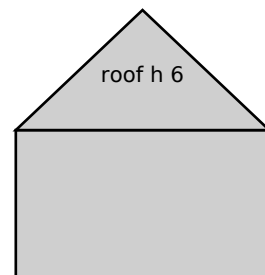


folded part: base 6, h 8

Working:

Answer:

4. A composite figure is made of a rectangle 16 cm by 12 cm and a triangular roof with base 16 cm and height 6 cm. Find the total area.



16 x 12

Working:

Answer:

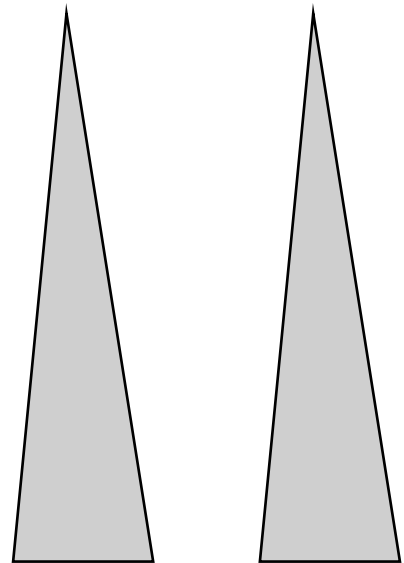
Set 2 - Shaded Areas

Level: Medium

Name: _____ Class: _____ Date: _____

Page 3

5. A rectangle is split into 3 identical triangles. Each triangle has base 10 cm and height 12 cm. Two triangles are shaded. Find the shaded area.

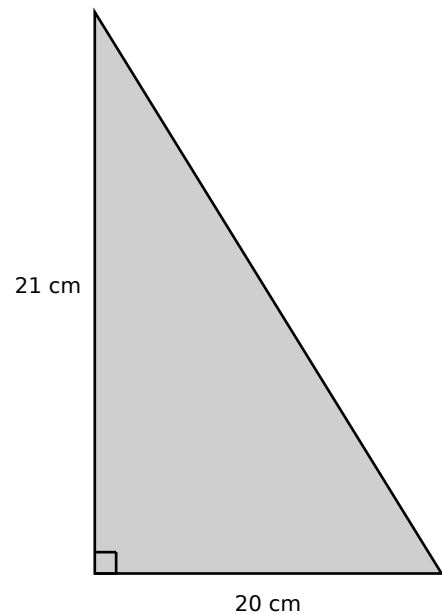


each: base 10 cm, height 12 cm

Working:

Answer:

6. A right-angled triangle has perpendicular sides 20 cm and 21 cm. Find its area.



Working:

Answer:

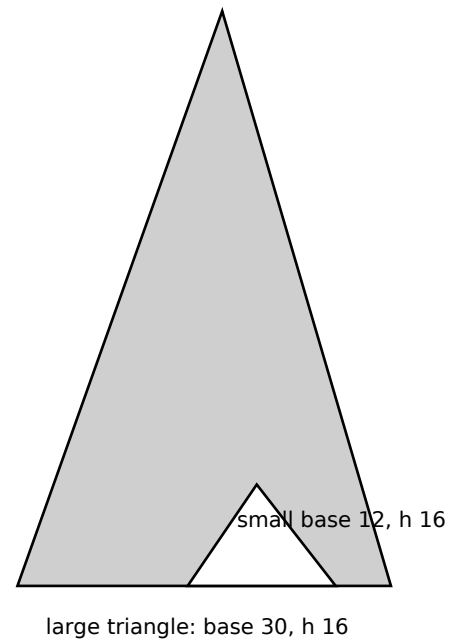
Set 2 - Shaded Areas

Level: Medium

Name: _____ Class: _____ Date: _____

Page 4

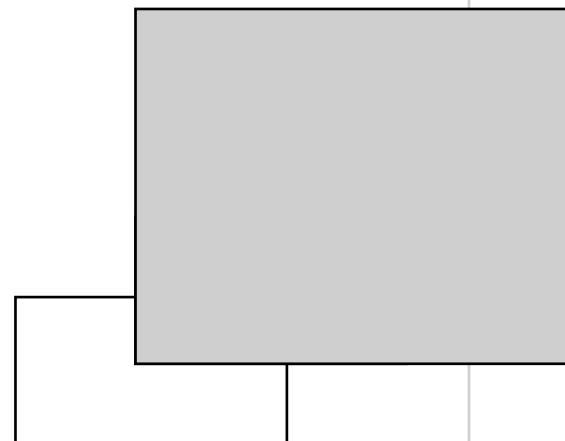
7. A large triangle has base 30 cm and height 16 cm. An unshaded triangle with the same height has base 12 cm. Find the shaded area.



Working:

Answer:

8. Two identical rectangles, each 16 cm by 7 cm, overlap. The overlapping area is 20 cm^2 . Find the area of the combined figure.



each 16 x 7; overlap 20

Working:

Answer:

Set 2 - Shaded Areas

Level: Medium

Name: _____ Class: _____ Date: _____

Page 5

- 9.** Ben had some stickers. He gave $\frac{2}{5}$ of them to his brother and $\frac{1}{3}$ of the remainder to his friend. He had 36 stickers left. How many stickers did he have at first?

Working:

Answer:

- 10.** Cara used $\frac{3}{8}$ of a ribbon. She then used $\frac{1}{5}$ of the remainder. She had 50 cm of ribbon left. How long was the ribbon at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	210 cm ²	$\frac{1}{2} \times 28 \times 15 = 210 \text{ cm}^2$
2	506 cm ²	Rectangle = 576; triangle = $\frac{1}{2} \times 20 \times 7 = 70$; shaded = 506 cm ²
3	24 cm ²	$\frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$
4	240 cm ²	Rectangle = 192; roof = $\frac{1}{2} \times 16 \times 6 = 48$; total = 240 cm ²
5	120 cm ²	Each = $\frac{1}{2} \times 10 \times 12 = 60$; 2 triangles = 120 cm ²
6	210 cm ²	$\frac{1}{2} \times 20 \times 21 = 210 \text{ cm}^2$
7	144 cm ²	Large = 240; unshaded = 96; shaded = 144 cm ²
8	204 cm ²	Total = $2 \times (16 \times 7) - 20 = 224 - 20 = 204 \text{ cm}^2$
9	90 stickers	Left = $\frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$ of total. $\frac{2}{5} = 36$, so total = 90
10	100 cm	Left = $\frac{5}{8} \times \frac{4}{5} = \frac{1}{2}$ of total. $\frac{1}{2} = 50$, so total = 100 cm

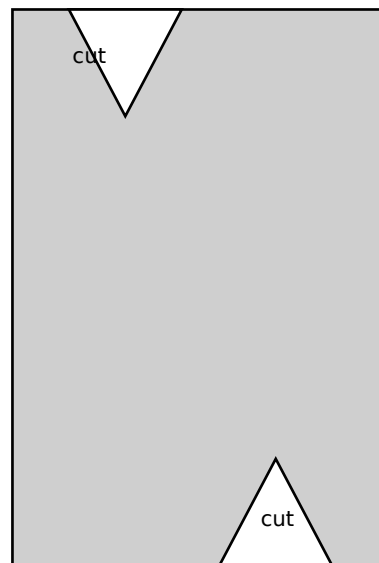
Set 3 - Composite Figures

Level: Medium to Hard

Name: _____ Class: _____ Date: _____

Page 1

1. A rectangle measures 36 cm by 20 cm. Two unshaded triangles are cut away. Each triangle has base 12 cm and height 8 cm. Find the shaded area.

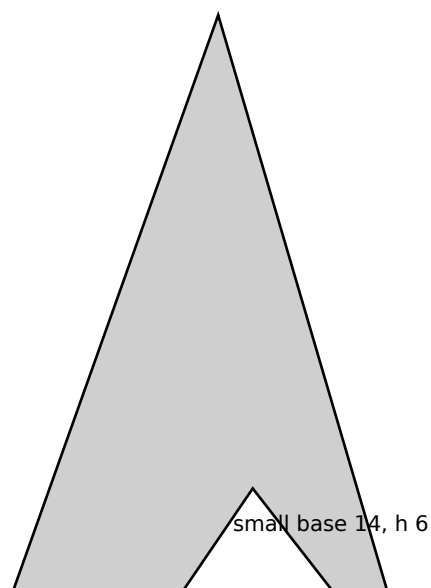


rectangle 36 x 20; each cut 12 x 8

Working:

Answer:

2. In Triangle ABC, BC = 42 cm and the height from A to BC is 18 cm. If the unshaded triangle has base 14 cm and height 6 cm, find the shaded area.



large triangle: base 42, h 18

Working:

Answer:

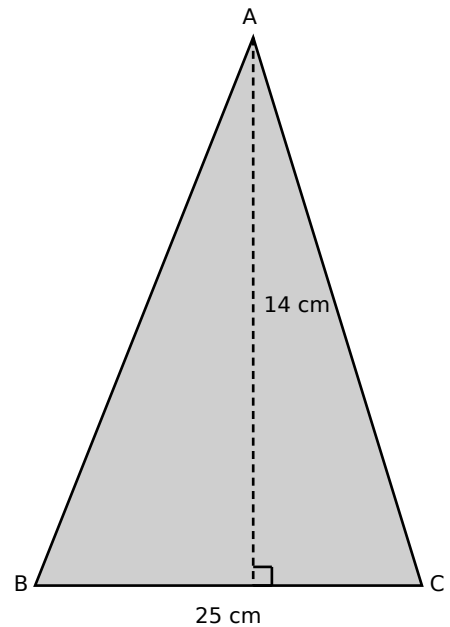
Set 3 - Composite Figures

Level: Medium to Hard

Name: _____ Class: _____ Date: _____

Page 2

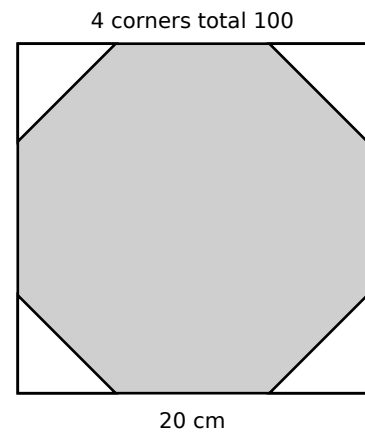
3. A triangle has base 25 cm and height 14 cm. Find its area.



Working:

Answer:

4. A square has side 20 cm. Four unshaded corner triangles have a total area of 100 cm^2 . Find the shaded area.



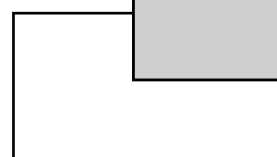
Working:

Answer:

Set 3 - Composite Figures

Name: _____ Class: _____ Date: _____

5. Two identical right-angled triangles each have area 60 cm^2 . They overlap by 18 cm^2 . Find the area covered by the two triangles together.

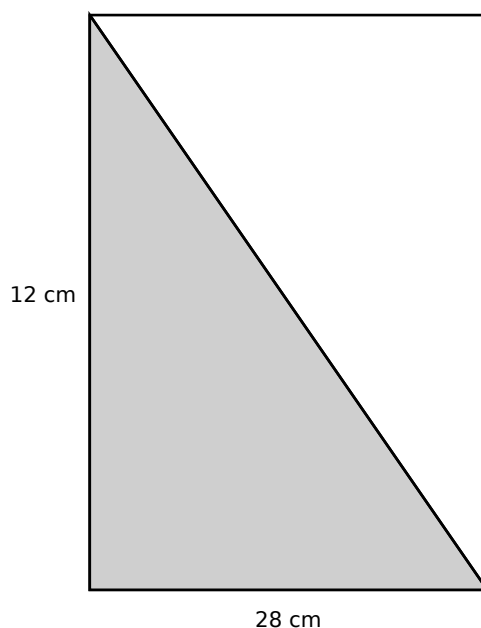


2 shapes, overlap 18

Working:

Answer:

6. In Rectangle ABCD, $AB = 28 \text{ cm}$ and $AD = 12 \text{ cm}$. Point E is on DC such that $EC = 18 \text{ cm}$. Find the area of shaded Triangle AEC.



Working:

Answer: _____

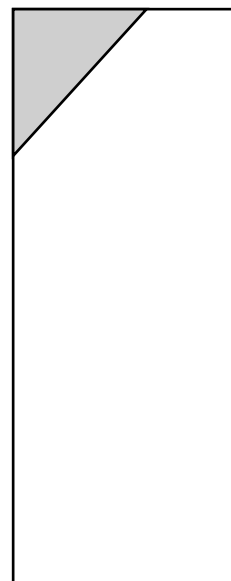
Set 3 - Composite Figures

Level: Medium to Hard

Name: _____ Class: _____ Date: _____

Page 4

7. A rectangle measuring 24 cm by 10 cm has a triangular corner folded over. The folded triangle has base 10 cm and height 8 cm. Find the remaining visible area.

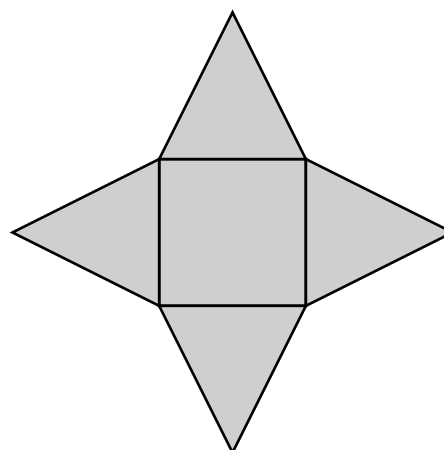


rect 24 x 10; fold 10 x 8

Working:

Answer:

8. A figure is made up of a square of area 64 cm^2 and four identical right-angled triangles. Each triangle has base 8 cm and height 6 cm. Find the total area.



square area 64 + 4 triangles

Working:

Answer: _____

Set 3 - Composite Figures

Level: Medium to Hard

Name: _____ Class: _____ Date: _____

Page 5

- 9.** Rina had some marbles. She gave $\frac{1}{6}$ of them to a friend and $\frac{2}{5}$ of the remainder to her cousin. She had 40 marbles left. How many marbles did she have at first?

Working:

Answer:

- 10.** Ali spent $\frac{3}{7}$ of his money on books and $\frac{1}{4}$ of the remainder on files. He had \$90 left. How much money did he have at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	624 cm ²	Rectangle = 720; two triangles = $2 \times 48 = 96$; shaded = 624 cm ²
2	336 cm ²	Large = $\frac{1}{2} \times 42 \times 18 = 378$; small = $\frac{1}{2} \times 14 \times 6 = 42$; shaded = 336 cm ²
3	175 cm ²	$\frac{1}{2} \times 25 \times 14 = 175$ cm ²
4	300 cm ²	Square = $20 \times 20 = 400$; shaded = $400 - 100 = 300$ cm ²
5	102 cm ²	Combined area = $60 + 60 - 18 = 102$ cm ²
6	108 cm ²	Base EC = 18, height = 12. Area = $\frac{1}{2} \times 18 \times 12 = 108$ cm ²
7	200 cm ²	Rectangle = 240; folded triangle = 40; remaining = 200 cm ²
8	160 cm ²	Each triangle = 24; 4 triangles = 96; total = $64 + 96 = 160$ cm ²
9	80 marbles	Left = $\frac{5}{6} \times \frac{3}{5} = \frac{1}{2}$ of total. Total = 80
10	\$210	Left = $\frac{4}{7} \times \frac{3}{4} = \frac{3}{7}$ of total. $\frac{3}{7} = 90$, so total = 210

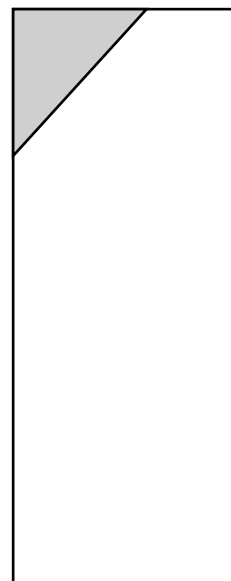
Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

Page 1

1. A rectangular paper measures 48 cm by 22 cm. Two corner triangles are folded in. Triangle Q has area 36 cm^2 and Triangle P has twice the area of Triangle Q. Find the shaded area left.

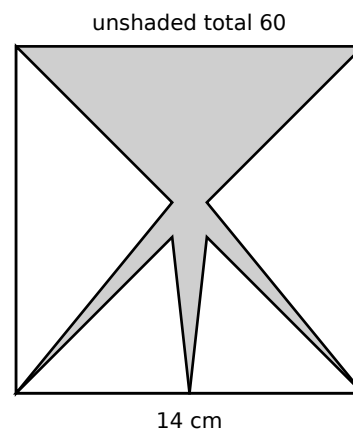


rectangle 48×22 ; $P = 2Q$; $Q = 36$

Working:

Answer:

2. A square PQRS has side 14 cm. The total area of four unshaded corner triangles is 60 cm^2 . Find the shaded area.



Working:

Answer:

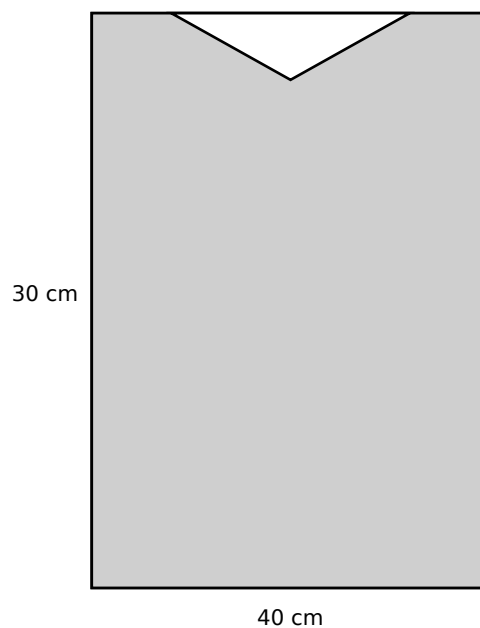
Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

Page 2

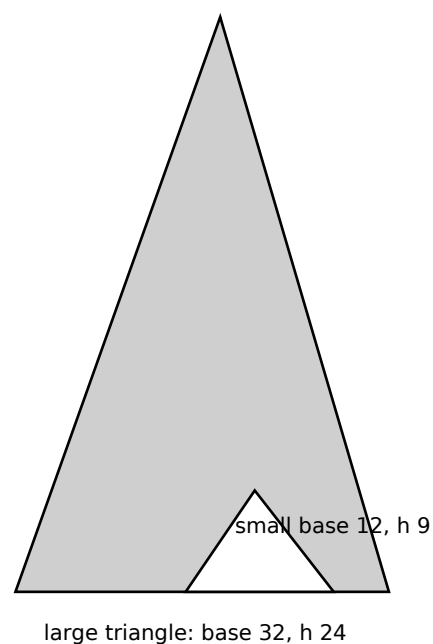
3. A rectangle measures 40 cm by 30 cm. A triangular part with base 18 cm and height 20 cm is unshaded. Find the shaded area.



Working:

Answer:

4. A large right-angled triangle has base 32 cm and height 24 cm. A smaller unshaded triangle has base 12 cm and height 9 cm. Find the shaded area.



Working:

Answer:

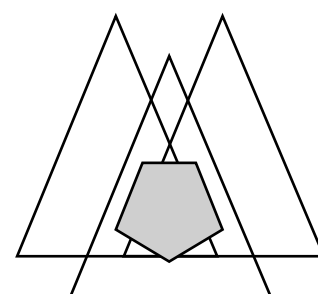
Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

Page 3

5. A pattern is made of 5 identical triangles. Each triangle has base 12 cm and height 10 cm. Three triangles are shaded. Find the shaded area.

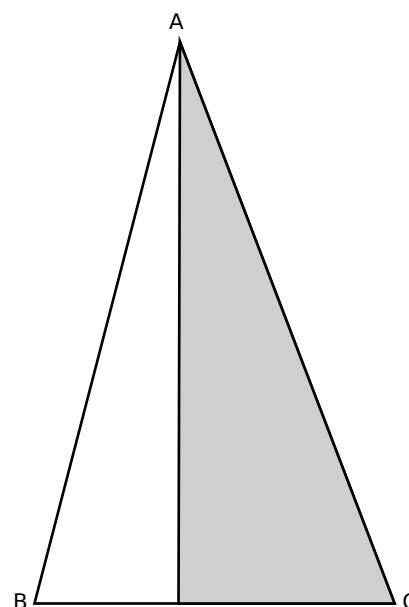


5 identical triangles; 3 shaded

Working:

Answer:

6. In Triangle ABC, BC is divided at D in the ratio $BD:DC = 2:3$. The area of Triangle ABC is 240 cm^2 . Find the area of Triangle ADC.



BC split 2:3
same height

Working:

Answer:

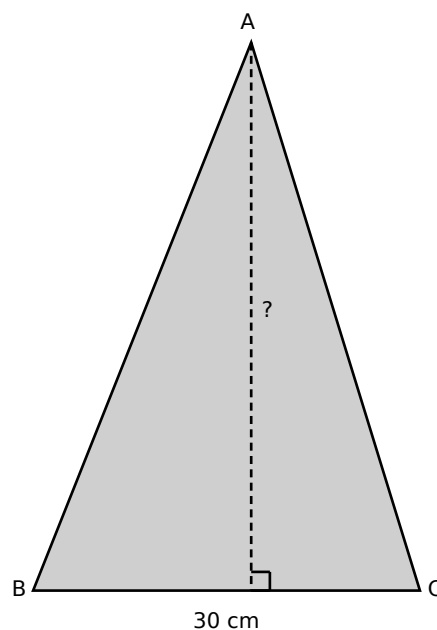
Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

Page 4

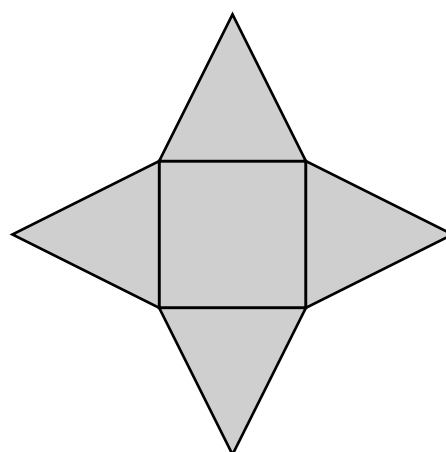
7. Triangle AEB has base $AB = 30$ cm and area 135 cm^2 . Find the perpendicular height from E to AB.



Working:

Answer:

8. A figure is made up of a square and 4 identical triangles. The square has side 10 cm. Each triangle has base 10 cm and height 8 cm. Find the total area.



square side 10 + 4 triangles

Working:

Answer:

Set 4 - Challenge Practice

Level: Hard

Name: _____ Class: _____ Date: _____

Page 5

- 9.** Lina donated $\frac{3}{8}$ of her books and sold $\frac{2}{5}$ of the remainder. She had 45 books left. How many books did she have at first?

Working:

Answer:

- 10.** A tank had some water. $\frac{1}{3}$ was used first, then $\frac{1}{4}$ of the remainder, then $\frac{2}{5}$ of the new remainder. If 24 litres were left, how much water was in the tank at first?

Working:

Answer:

Set 4 - Answer Sheet

Level: Hard

Name: _____ Class: _____ Date: _____

Page 6

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	948 cm ²	Rectangle = 1056. Q = 36, P = 72. Shaded = 1056 - 108 = 948 cm ²
2	136 cm ²	Square = 196; shaded = 196 - 60 = 136 cm ²
3	1020 cm ²	Rectangle = 1200; triangle = $\frac{1}{2} \times 18 \times 20 = 180$; shaded = 1020 cm ²
4	330 cm ²	Large = 384; small = 54; shaded = 330 cm ²
5	180 cm ²	Each triangle = $\frac{1}{2} \times 12 \times 10 = 60$; 3 shaded = 180 cm ²
6	144 cm ²	DC is $\frac{3}{5}$ of BC. Area ADC = $\frac{3}{5} \times 240 = 144$ cm ²
7	9 cm	$135 = \frac{1}{2} \times 30 \times h$, so $h = 270 / 30 = 9$ cm
8	260 cm ²	Square = 100; each triangle = 40; 4 triangles = 160; total = 260 cm ²
9	120 books	Left = $\frac{5}{8} \times \frac{3}{5} = \frac{3}{8}$ of total. $\frac{3}{8} = 45$, total = 120
10	80 litres	Left = $\frac{2}{3} \times \frac{3}{4} \times \frac{3}{5} = \frac{3}{10}$ of total. $\frac{3}{10} = 24$, total = 80

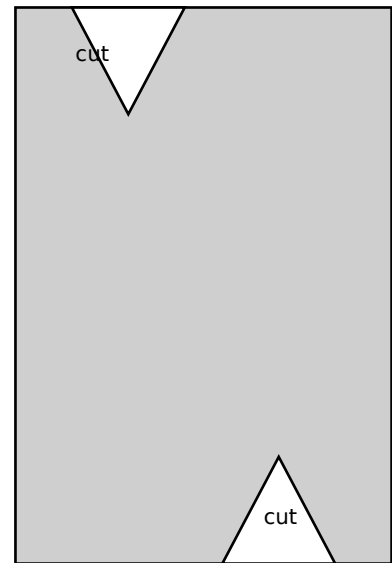
Set 5 - Mixed Exam-style

Level: Mixed Challenge

Name: _____ Class: _____ Date: _____

Page 1

1. Sharon has a rectangular paper measuring 60 cm by 30 cm. She cuts five similar triangles from the top. Each triangle has base 12 cm and height 15 cm. Find the area of the remaining paper.

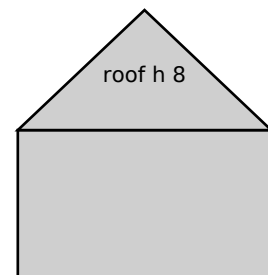


60 x 30; 5 cuts, each 12 x 15

Working:

Answer:

2. A house-shaped figure is made of a rectangle 18 cm by 14 cm and a triangular roof with base 18 cm and height 8 cm. Find the total area.

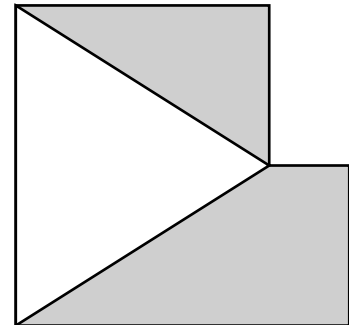


18 x 14

Working:

Answer:

3. An L-shaped figure is made from a 36 cm by 24 cm rectangle by removing a 12 cm by 8 cm rectangle. Find the area of the L-shaped figure.

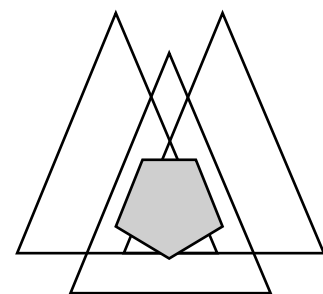


36 x 24 minus 12 x 8

Working:

Answer:

4. Three identical triangles overlap. Each triangle has base 30 cm and height 20 cm. The central shaded overlap has area 75 cm^2 and is counted in all 3 triangles. Find the area covered by the whole figure.

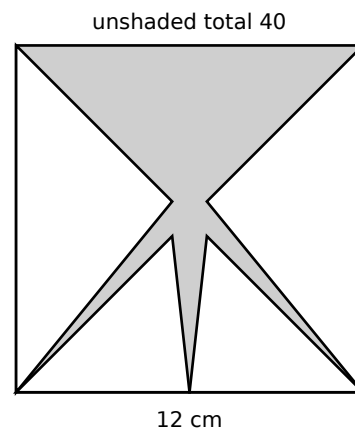


3 triangles; overlap 75

Working:

Answer: _____

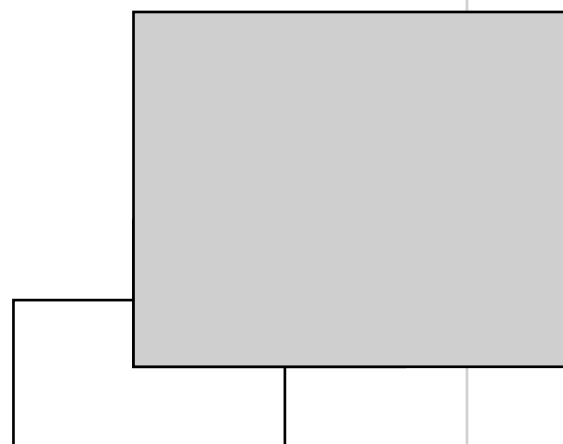
5. A square paper measures 12 cm by 12 cm. The total area of the unshaded triangles is 40 cm². Find the shaded area.



Working:

Answer:

6. Two identical rectangles each measure 20 cm by 8 cm. They overlap by 48 cm². Find the area of the combined figure.

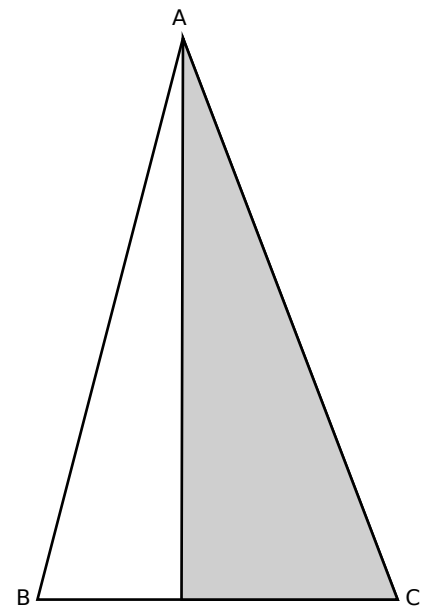


Working:

Answer:

Name: _____ Class: _____ Date: _____

7. In Triangle ABC, D lies on BC. $BD = 10$ cm and $DC = 8$ cm. The area of Triangle ADC is 32 cm^2 . Find the area of Triangle ABC.

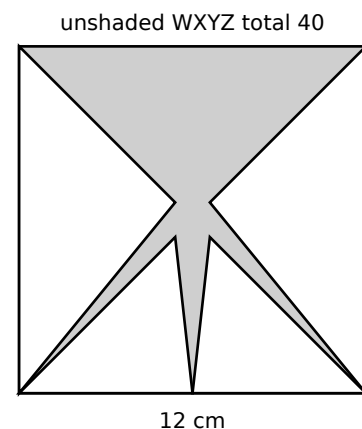


$BD=10$, $DC=8$
same height

Working:

Answer: _____

8. PQRS is a square with side 12 cm. The total area of the unshaded triangles W, X, Y and Z is 40 cm^2 . Find the area of the shaded part.



Working:

Answer: _____

Set 5 - Mixed Exam-style

Level: Mixed Challenge

Name: _____ Class: _____ Date: _____

Page 5

- 9.** Ethan spent $\frac{2}{7}$ of his money on a bag and $\frac{3}{5}$ of the remainder on shoes. He had \$80 left. How much money did he have at first?

Working:

Answer:

- 10.** Zoe baked some muffins. She gave $\frac{1}{4}$ of them to her neighbours and sold $\frac{1}{3}$ of the remainder. She had 30 muffins left. How many muffins did she bake at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	1350 cm ²	Rectangle = 1800. One triangle = 90. Five = 450. Remaining = 1350 cm ²
2	324 cm ²	Rectangle = 252; roof = $\frac{1}{2} \times 18 \times 8 = 72$; total = 324 cm ²
3	768 cm ²	$36 \times 24 - 12 \times 8 = 864 - 96 = 768 \text{ cm}^2$
4	750 cm ²	Total counted = $3 \times 300 = 900$. Overlap counted 3 times but needed once, subtract 2×75 Area = 750 cm ²
5	104 cm ²	Square area = 144; shaded = $144 - 40 = 104 \text{ cm}^2$
6	272 cm ²	$2 \times 160 - 48 = 272 \text{ cm}^2$
7	72 cm ²	Same height. $BC/DC = 18/8$. Area ABC = $32 \times 18/8 = 72 \text{ cm}^2$
8	104 cm ²	Square = 144; shaded = $144 - 40 = 104 \text{ cm}^2$
9	\$280	Left = $\frac{5}{7} \times \frac{2}{5} = \frac{2}{7}$ of total. $\frac{2}{7} = 80$, total = 280
10	60 muffins	Left = $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ of total. Total = 60